

From Agile Software Development to Agile Businesses

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Abstract

Agile software development has proven to be a good fit for knowledge work in environments of high change rates, often resulting in hyper-productive teams. The bottom line seems to be that applying agile methods to software development projects reduces risk and time-to-market while high quality is achieved at the same time. We show how agility can be useful well beyond the context of individual projects, first when managing a portfolio of projects of a company and second when maneuvering a company in the marketplace. Companies can become more demand-driven and value-driven by embracing agility on these three levels and benefit from agility over a wider spectrum. Agile businesses react fast on changed circumstances, are flexible, bring response to the market as fast as possible, are proactive, and sense the change.

1. Agility on three levels

Agile methods have been used on software development projects successfully for a couple of years. Going back in history, iterative and incremental methods have already been used successfully for a couple of decades (see [17]).

When something works well in a particular context, it may be worthwhile to apply it to the next bigger context, as well. The next bigger context to software development projects is the portfolio of projects that a company undertakes or plans to undertake. The yet next bigger context is the entire enterprise where the company is challenged by its competitors (see Fig. 1).

At the project level, agile engineering and project management practices are well established and help to meet the project's targets such as budget, schedule, functionality, and quality. At the portfolio of projects level, governance structures are used to align the projects with the business goals and to continuously assess and transform the portfolio. Individual projects undergo financial analysis and are evaluated against each other by means of return on investment (ROI),

total cost of ownership (TCO), internal rate of return (IRR), net present value (NPV), payback period and break even. Techniques such as real options may be used if the environment is too uncertain to predict. At the enterprise level, mergers and acquisitions have once been a way to rapidly grow a company, but the resulting company may lose its focus and be less responsive afterwards. Other approaches may be more appropriate in the current economic situation. Companies establish vision statements and intellectual agendas, and derive their business strategy and IT strategy from these.

These three levels are connected via iterations of {Assess – Plan – Develop Guidance / Perform – Measure} where the third step leads down to the next lower level.

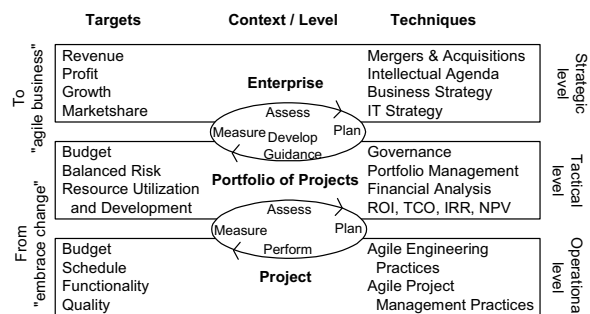


Figure 1: The three levels with their targets and techniques

For the three levels (single project, portfolio of projects, enterprise), agility brings similar yet different benefits, while agility changes its meaning from simply embracing change on the project level to business agility on the enterprise level. We see two approaches for companies to become agile across all three levels: "bottom up" by growing agile techniques that have been applied at the project level to wider parts of the business, and "top down" by defining a new vision and strategy for the business and then breaking it down from the strategic level to the tactical and operational levels.

In the rest of the article, we detail what has been achieved and what looks promising with respect to agility on the different levels. We continue with perspectives on IBM's approach and give examples of solutions and offerings. Finally, we summarize the benefits of agility on these various levels.

2. Agility at the project level

Agile methods have traditionally had a strong focus on engineering practices for the development team and on project management practices. These practices have recently been enhanced by several financial ideas such as real options. Together these practices and the methodological guidance are very effective for typical software development projects with uncertain or changing business requirements and with technological uncertainties.

2.1. Agile methods from a technical and project management point of view

Agile methods like Scrum [23], XP [5], DSDM [11], Crystal [9], etc. cover all aspects of delivering software in an agile way. These methods are based on research and experience from Lean Manufacturing [26], from Lean Software Development [21] and from Knowledge Management [19].

2.2. Agile methods from a financial point of view

In the early days of agile methods, the focus has been on engineering aspects: emergence of requirements, architecture, design, etc. Maybe as a consequence of early and frequent delivery of software, another property of agility emerged: financial flexibility and options. Traditionally, a business case for a project contained a time series with the inflow and outflow of cash, the discounted cash flows (DCF), the net present value (NPV), the return on investment (ROI), and internal rate of return (IRR). For agile projects, with their frequent delivery of running software, the idea of real options (as an analogy to financial options, see e.g. [13]) came to the mind of agile thought-leaders: When you have a "call option" for a certain stock, you have the right to buy that stock for a pre-determined price. You pay for this flexibility, since it can pay off if the stock price rises. In a similar way, you pay for a "real option" in order to have the flexibility to use it to your favor. Such a real option may be the first iteration in a software development project where you do not yet know how the market will respond to your new software. If the market likes your

idea, you are in a better position to proceed. The idea behind real options is used to value various possible approaches while there are still a lot of uncertainties.

Another idea has been pursued in [10] with the "Incremental Funding Method": a way to break down a project's requirements into "minimum marketable features" (MMFs, as units of software value creation) with intrinsic marketable value (e.g. competitive differentiation, revenue generation, cost saving, brand projection, enhanced loyalty, etc.) and into "architectural elements" (AEs) which have cost and duration but no associated revenues. In a next step, the dependencies among the MMFs and AEs are identified, and the MMFs are evaluated (considering their tangible and intangible benefits, quantifying the value in terms of cost and revenue projections, taking into account their risks and duration), discounting the calculated values to present values, and determining financially attractive sequences for delivering the MMFs and AEs.

This last step is where IFM differs greatly from previous agile methods:

- XP uses a greedy approach to deliver the features first that have the highest value for the client. This can lead to a sub-optimal return for the entire project due to local optimization.
- RUP [16] uses a risk-driven approach to deliver the features first that have the highest risk. This can – as well – lead to a sub-optimal return for the entire project, if the riskiest parts are not needed in the end.
- IFM uses heuristics that look ahead to evaluate sequences of MMFs and AEs. This leads to the architecture being delivered incrementally, just as much as necessary to get optimal business value out of a project with commensurate investment.

2.3. Benefits of agility on the project level

At the level of a single project, the benefits are:

- reduced risk by involvement of user and customer (with early feedback and regular synchronization)
- reduced time-to-market by early and regular delivery of functionality
- increased productivity by focusing on value-adding features and value-creating activities
- empowered team
- increased quality, software with perceived and conceptual integrity

3. Agility at the portfolio of projects level

Let us next have a look at traditional portfolio management and then argue how agility comes to play.

3.1. Traditional portfolio management

Within the scope of IT, a portfolio can be defined as the set of current and future components (which can be hardware, software, or services offerings, solutions, or projects, etc.) within a business area that addresses a market segment. Portfolio management can be defined as the collection, assessment, and management of a portfolio against selected business criteria. In the following, we shortly describe the typical activities:

- **Defining the portfolio:** understand the business strategy; identify the portfolio sponsor; define the business decision criteria (revenue / benefits, costs, gross profit, etc.) with linkage to the business strategy (used to select components and to track & assess portfolio performance); identify initial risks (including component dependencies and resource capacity/demands)
- **Planning the portfolio:** establish the portfolio management; develop the inventory of components and the resources to be managed; select and prioritize the components
- **Collecting/Improving the portfolio:** collect (qualitative and quantitative) portfolio information; assess the portfolio components (health and contribution indicators, resource demands, risks); manage the portfolio (focusing on non-compliance with business objectives, basing actions on impact to overall portfolio – e.g. reallocate resources, cancel projects, sunset products); improving the portfolio (incorporate Lessons Learned); review the portfolio; assess portfolio management maturity

The benefits of portfolio management are increased executive governance and a holistic approach: to ensure the portfolio continues to align with the business strategies, to provide a balanced portfolio (risk and value) within the organization's capabilities and constraints, to validate that the business decision criteria established were accurate and that the expected results are realized, to ensure resource capacity can meet the demand, to take into account all the relationships between the components within a portfolio (direct dependencies, indirect dependencies, common sub-components), and to make decisions that are in the best interest of the overall portfolio strategy even if they are detrimental to any specific component, or favor one component over the other.

3.2. Agile portfolio management

We take the ideas from Lean Manufacturing and Lean Software Development as a guideline for indicating areas that can benefit from agility within portfolio

management (the items in the following list are the lean principles from [21] followed by the agile tools in the brackets):

- **Eliminate waste** (Seeing waste, Value stream mapping): At the level of portfolio management, projects that are not aligned with the business strategy are waste. The process of defining, planning, and improving a portfolio needs to be stream-lined, e.g. unnecessary delays, inventory, etc. need to be identified and eliminated.
- **Amplify learning** (Feedback, Iterations, Synchronization, Set-based development): Rapid feedback based on actual data allows for fact-based decision making, synchronization of people from different levels (higher management, lower management, delivery) on the business strategy is crucial to continued alignment of IT with business. Synchronization within the team of portfolio analysts should happen very often (e.g. daily), synchronization of the analysts with the executives and the project team leaders should happen regularly (e.g. weekly to monthly). It is important to keep the feedback cycle short by breaking down the initiatives into small enough projects.
- **Decide as late as possible** (Options thinking, The last responsible moment, Making decisions): Individual portfolio components are no longer judged in separation but within the context of the entire portfolio. It is easy to understand the dependencies of portfolio components on each other and the impact of one on the other if the actual data about them is automatically collected and aggregated. Playing around with options allows to develop future scenarios and to simulate possible outcomes without long delays. Decisions can be delayed as long as possible, but be made as soon as actual experience allows for fact-based decisions.
- **Deliver as fast as possible** (Pull systems, Queuing theory, Cost of delay): The cost of delay can be considered not just for one project, but for a sequence or web of projects. An understanding of the critical path for realizing business benefits over several projects and frequent inspection can be used to adapt to changes in a holistic way, i.e. not by sub-optimizing a particular project, but by optimizing the portfolio of projects.
- **Empower the team** (Self-determination, Motivation, Leadership, Expertise): With a shared understanding of the business strategy, middle management can work on breaking down the high-level business strategy to tactical guidance for the projects, and project team members can intuitively align the project priorities with the business

strategy. Motivation increases if employees understand the bigger context. In agile software development, there is no longer a single person responsible for activities like planning, estimating, etc. Rather the entire team accepts the responsibility. For managing a portfolio, one can try to move the power of planning and monitoring the portfolio down one level from the portfolio analysts to the project team leaders by sharing the available information about the projects. This can lead to a much better understanding and to fewer battles amongst the project teams.

- **Build integrity in** (Perceived Integrity, Conceptual integrity, Refactoring, Testing): Individual projects can be refactored (i.e. changing the delivery approach, changing the assigned resources, etc. while maintaining the externally visible results of the project, such as scope, cost, quality, etc.) to ameliorate the risk profile of the project. This can also be used to help troubled projects even at the expense of non-troubled projects, i.e. as a consequence of the refactoring, the previously non-troubled project acts e.g. as a donor of highly-skilled resources and suffers in its externally visible results to benefit the entire portfolio of projects.
- **See the whole** (Measurements, Contracts): The holistic approach helps to avoid sub-optimization. Albert Einstein once said “Not everything that can be counted counts, and not everything that counts can be counted.” Start with a small set of measurements, collect the data and verify whether the measurements help to manage the portfolio, use short feedback loops to adapt the measurement system to the overall goals. On the portfolio of projects level, techniques like Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT) can be used more effectively to understand and manage the sequence or the web of projects than on the project level to manage the internals of projects.

3.3. Benefits of agility on the portfolio of projects level

At the portfolio level, the benefits are:

- reduced overall risk by managing a portfolio of projects, some with more risk, some with less
- re-positioning with respect to the competitors during the course of the years
- reduced complexity (e.g. due to a reduced number of applications, projects, etc. and due to removed duplicates) and reduced maintenance efforts

- optimized utilization and development of the resources
- understanding of the gaps between the current state and the desired state
- improved responsiveness, flexibility, and productivity

4. Agility at the enterprise level

For companies, to be more agile means to be more responsive, focused, variable, and resilient (see [20]). Becoming an agile business is not something that can happen all at once and will surely take an extended amount of time. In an environment of changes, all enterprise-wide undertakings should choose a “breadth-first” approach (where one first gets an overview, and then only selectively drills down to the details of the relevant parts) rather than a “big bang” approach (where one tries to get a detailed view of everything from the beginning, i.e. waterfall thinking).

4.1. Analytical top down approach

In the “top down” approach of becoming an agile business, the business takes the lead, and governance as well as continued alignment assures that the IT supports the business in the long term. Strategic initiatives may first break down the business into “business components” (logical grouping of the people, technology and resources that delivers specific business value and can operate independently) and then setup rules, principles and standards for the future business and IT architectures.

Goals of strategic initiatives may be to:

- show how well aligned the organizational capabilities are with the strategy
- show how strongly the investments are aligned with the strategy
- create a step-by-step integrated business and technology roadmap to address key issues and pain points
- show where capabilities overlap on both the business and technology levels and provide a clear path to reconcile

When the business has been broken down into business components, one may assess whether the business components lead to a competitive advantage, and whether the revenue contribution and costs are well-aligned. Coase’s Law [8] says that firms should perform only those functions internally that cannot be performed more cheaply by the market. This can lead to the question of sourcing, whether to insource business components (where the company differentiates from its competitors) that were

previously outside the enterprise, and whether to outsource other business components. Between these extremes, there are several variants. On the sourcing level, agility means to be able to scale up and scale down with the demand.

4.2. Collaborative bottom up approach

General Electric has institutionalized a program called “Work-Out” to learn and improve as an organization. Work-Out consists of three stages (see [25]):

1. Plan: A senior business leader pulls together a small "design" team to agree on the key issues to be tackled, to identify participants, to communicate expectations of what people will do at the Work-Out event, and to prepare business leaders for their role.
2. Conduct: The workshop (2-3 days) pulls together people from all over the enterprise. The main objective is to maximize the number of high-quality ideas for change and to carve out a doable list of actions that will deliver fast results on the improvement opportunity. Ideas are generated in small groups. After presentation to the whole group, the fruitful ideas are developed further in small groups. At the end of the event, the ideas are presented to the business leader who decides about the idea on the spot.
3. Implement: make accountability stick, manage the organization's cycle of energy, translate ideas into action and provide on-going management oversight and support.

Work-Out has been very effective at eliminating waste, tearing down hierarchy and red tape, in creating an empowered workforce confident enough to envision and implement change whenever and wherever it is needed, changing the corporate culture to be more agile.

4.3. Benefits of agility on the enterprise level

At the enterprise level, the benefits are:

- becoming more responsive, focused, variable, and resilient
- integrating and optimizing across the value net and value chain
- right-sourced business components (from insourcing to outsourcing)

Companies can gain flexibility by considering various sourcing models, from simply augmenting the in-house staff with special resources to full outsourcing with transferring employees.

5. Perspectives on IBM's approach

In the following, we try to give our view on IBM's approach. We can only speak from our background and experience and will surely miss several offerings, tools and approaches (due to the size and diversity of IBM).

5.1. On demand business

In 2002, Sam Palmisano, Chairman of the Board and CEO of IBM, gave his definition of an on demand business as part of IBM's new intellectual agenda: “An enterprise whose business processes – integrated end-to-end across the company and with key partners, suppliers and customers – can respond with flexibility and speed to any customer demand, market opportunity or external threat.” [20]

The ever-changing business environment with competitive Darwinism (i.e. survival of the fittest), continuous discontinuities, unrelenting financial pressures and unpredictable threats drives businesses to be more focused, responsive, variable, and resilient (from [14]):

- **Competitive Darwinism:** Due to increasingly transparent markets, competitive intensity is growing in severity as existing players and new entrants scramble to provide value to customers. Therefore, being **focused** means being committed to concentrating on differentiating competencies, using tightly integrated strategic partners to manage selected non-differentiating activities.
- **Continuous discontinuities:** The changes in customer demands, technological innovations and government regulations are increasingly sudden and dramatic. Therefore, being **responsive** means seeming almost intuitive in one's ability to sense and respond rapidly to unpredictable changes in the market environment and the needs of all its constituents.
- **Unrelenting financial pressure:** The growth and predictability of revenues and margins becomes even more challenging due to economic uncertainties and demanding investors. Therefore, being **variable** means being able to adapt cost structures and business processes flexibly, in order to reduce risk and to do business at higher levels of productivity, cost control, capital efficiency and financial predictability.
- **Unpredictable threats:** Pervasive dangers become real and prevalent as global firms see increased exposure to natural disasters, unstable geopolitics and other market shocks. Therefore, being **resilient** means being prepared for changes

and threats – be they technological, economic or political – enabling the business to continue operating with consistent availability, security and privacy.

5.2. Offerings, Tools and Sourcing

In the recent years, IBM has created several offerings on the various levels and gained extensive practical experience helping the clients on their way to on demand businesses. The following figure only represents our personal view. There may well be additional offerings, tools and sourcing variants that we are not aware of. We discuss the offerings in the following sub-sections and give pointers for additional information.

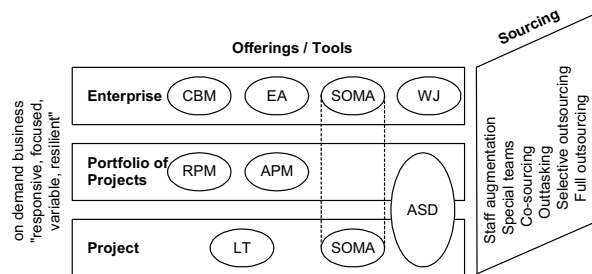


Figure 2. Offerings, tools and sourcing

5.2.1. IBM's Component Business Modeling (CBM). CBM [4] is a technique for modeling an enterprise into non-overlapping and unique business components. Each business component is a logical grouping of the people, technology and resources that deliver specific business value and can operate independently. CBM creates a map of the essential "building blocks" of an organization into a tabular structure that can easily fit on a single page. One can then evaluate the business components for various properties and identify the components with the highest pressure for modification.

5.2.2. IBM's Enterprise Architecture (EA). EA [12] develops a description of how a business should be structured, in order to implement and support the business strategy of the organization. As such, the business architecture provides a bridge between the business strategy and the detailed definition of the main resources (process, knowledge, information, organization and technology) required to implement it. EA also defines a high level enterprise-wide IT Architecture which normally consists of three component parts: an Application Architecture, a Data Architecture and a Technical (Infrastructure) Architecture. Additionally, EA performs a strategic gap

analysis (current state vs. future state) and sets up a governance model in order to facilitate the transition.

5.2.3. IBM's Service-Oriented Modeling and Architecture (SOMA). SOMA [24] helps companies to implement a service-oriented architecture (SOA) with techniques for service identification (e.g. domain decomposition, goal-service modeling, existing system analysis), service specification and service realization (see [3]).

5.2.4. IBM's World Jam (WJ). WJ [27] engages all employees in a collaborative discussion forum for 72 hours via IBM's intranet. The idea is similar to GE's Work-Out but implemented via the intranet. The difference is the distributed setting of the global community via the intranet. After the world jam sessions, the ideas are then consolidated. The critical point (when comparing with GE's Work-Out), is to ensure that the ideas are acted upon and then to communicate the actions and results.

5.2.5. IBM's Rational Portfolio Manager (RPM). RPM [22] is an enterprise project management tool that covers all aspects of project management (planning, staffing, tracking, financials, collaboration, exceptions and reporting) for small, medium and large projects. It has extensive support for the management of portfolios (from an executive point of view) with OLAP functionality: It aggregates the data from various projects to the portfolio of projects view, presents the aggregated data in form of portfolio dashboards (reports, pivot tables, scorecards, bubble charts) and then allows to again drill-down to the underlying data. It collects and reports on current and historical project execution status and provides trends.

5.2.6. IBM's Application Portfolio Management (APM). APM [1] is an offering to continuously assess, transform and manage a client's portfolio of applications to meet business needs and objectives. Management of application portfolios is divided into portfolio assessment and portfolio transformation. Typical recommendations at the end of the assessment are: retire, replace, restructure, reprioritize, or relocate an application. The assessment is embedded in a continuous cycle of {Assess – Plan – Transform – Measure}. The later parts of this cycle plans the transformation projects, performs them (which leads us down to the project level with agile software development or legacy transformation projects), and measures their results and their effect on the portfolio and enterprise.

5.2.7. IBM's Accelerated Solution Delivery (ASD).

ASD [2] is an example of large-scale co-sourcing. Since a couple of years, IBM has installed Accelerated Solution Delivery Centers with various clients. In these centers, shared teams (50% IBM, 50% client personnel) work in an agile way, with time-boxing, prioritized features, joint application design, co-located small teams of 4-8 persons, metrics to measure the improvements etc. A couple of teams (in some cases up to 80) work in parallel and synchronize on a regular basis. The results are typically a reduction of time-to-market by up to 40%, a reduction of development costs by up to 30%, a significant reduction of defects, a significant increase in productivity and a reduction of employee attrition. In such centers, IBM initially brings in experience with implementing the ASD model and experienced resources; whereas the customer brings in the business knowledge. After a couple of years, the client can take over the entire center.

5.2.8. IBM's Legacy Transformation (LT). LT [18] provides a full range of offerings that are designed to address common legacy-related challenges and costs while helping to incrementally position the IT environment for new initiatives such as e-business on demand. Used individually or together, these offerings are focused on improving the management and maintenance of the portfolio.

5.3. Evaluation to increasing the agility

From the quoted offerings, only ASD is inherently agile (i.e. when looking at the delivery approach). It gives convincing evidence that agile software development techniques can be used on a large scale.

The other offerings (CBM, EA, SOMA, APM, and LT) can be performed in an agile or in a more traditional plan-driven way. They all help the companies to become more focused, responsive, variable and resilient, even if the individual projects are not performed in an agile way. It is interesting to understand that the delivery approach of a project does not necessarily have to be agile (at the project level) to deliver agile benefits at a higher level. Nevertheless, it may be worthwhile to trim the delivery approaches from an agile point of view.

6. Conclusions

Agility brings different benefits depending on the context in which it is applied. A simple statement like "embrace change" (from the project level, to quote Kent Beck) then means that an agile business reacts

fast on changed circumstances, is flexible, brings response to the market as fast as possible, is proactive, and senses the change.

There seem to be two important facets of agility: the delivery approach and the realized results and benefits. On the project level, teams use an agile delivery approach to reach their target and to become hyper-productive (see [6]). On the portfolio of projects and enterprise level, the realized results and benefits are business agility with being more responsive, focused, variable, and resilient. It is well possible to reach agile results without applying an agile delivery approach as we have seen in section 5.3.

We have two main conclusions:

1. Simply using an agile approach on one level without considering the next bigger level, will not lead to the optimum return. One has to consider the next bigger context.
2. Although agile results may be realized without an agile delivery approach, it may be worthwhile to move to an agile delivery approach (as we have exemplified in section 3.2), since it may significantly increase the productivity, etc. (see section 2.3).

6.2. Inspiration and cross-fertilization

In addition to what we have laid out, it seems valuable to perform Benefits Realization Analysis on the portfolio of projects level as well and to integrate other ideas from [7], e.g. Value Realization Feedback into portfolio management.

It may as well be valuable to cross-fertilize across the different levels:

- to use techniques from agile software development for portfolio management, e.g. work in pairs, distribute the power, stop the production line and determine the root cause
- to use techniques from portfolio management for agile software development, e.g. apply rigorous financial analysis (with ROI, NPV, IRR, real options) not just to a portfolio of projects, but also to individual projects

6.3. Areas of future research

We do not expect a lot of radically new ideas on agile practices for software development and project management in the years to come but rather a consolidation of the best techniques and a clearer understanding for when (in which contexts and under which forces) to apply the techniques. We expect a lot of experience to be gained by using agile methods

appropriately to deliver the results on a consistent basis and on a larger scale.

When the delivery approach switches from plan-driven (focusing on efficiency and optimization) to agile (focusing on innovation and adaptability), this also means to switch from conformance to plan to conformance to vision and value, i.e. the targets on the project level may be rearranged (see [15]). We can not yet tell whether such a switch fits as naturally to the portfolio of projects level as it fits to the project level. When the delivery approaches have become agile on all three levels, the connection between the levels may no longer be as plan driven as now (“Assess – Plan – Develop Guidance / Perform – Measure”).

7. Acknowledgements

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